

5 (a) The contraction of striated muscle is explained by the sliding filament model.

- (i) Describe what happens in the sarcomere when the myosin head releases ADP and inorganic phosphate (Pi).

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- (ii) Explain the **precise** function of ATP in the sliding filament model.

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- (b) During contraction, muscles use up ATP very quickly. For a short period of time, ATP can be resynthesised using creatine phosphate, as shown in Fig. 5.1.

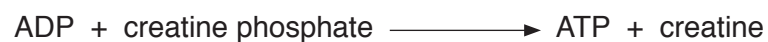


Fig. 5.1

The creatine formed as a result of the resynthesis of ATP is converted to creatinine. Creatinine production in the body stays fairly constant. Creatinine becomes part of the glomerular filtrate during ultrafiltration in the kidney nephrons.

- (i) Ultrafiltration requires a high blood pressure in the glomerulus.

Explain how this high blood pressure is achieved.

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- (ii) Name the main filtration barrier in the nephron that allows creatinine to pass into the renal capsule but stops red blood cells from passing through.

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- (c) The concentration of creatinine in the blood largely depends on the glomerular filtration rate (GFR). By measuring the concentration of creatinine in the blood, the GFR can therefore be estimated. The value of the GFR can be used to assess the efficiency of the kidneys.

In humans, a normal value of the GFR is $100\text{ cm}^3\text{ min}^{-1}$.

Fig. 5.2 shows the relationship between the GFR and the concentration of creatinine in the blood.

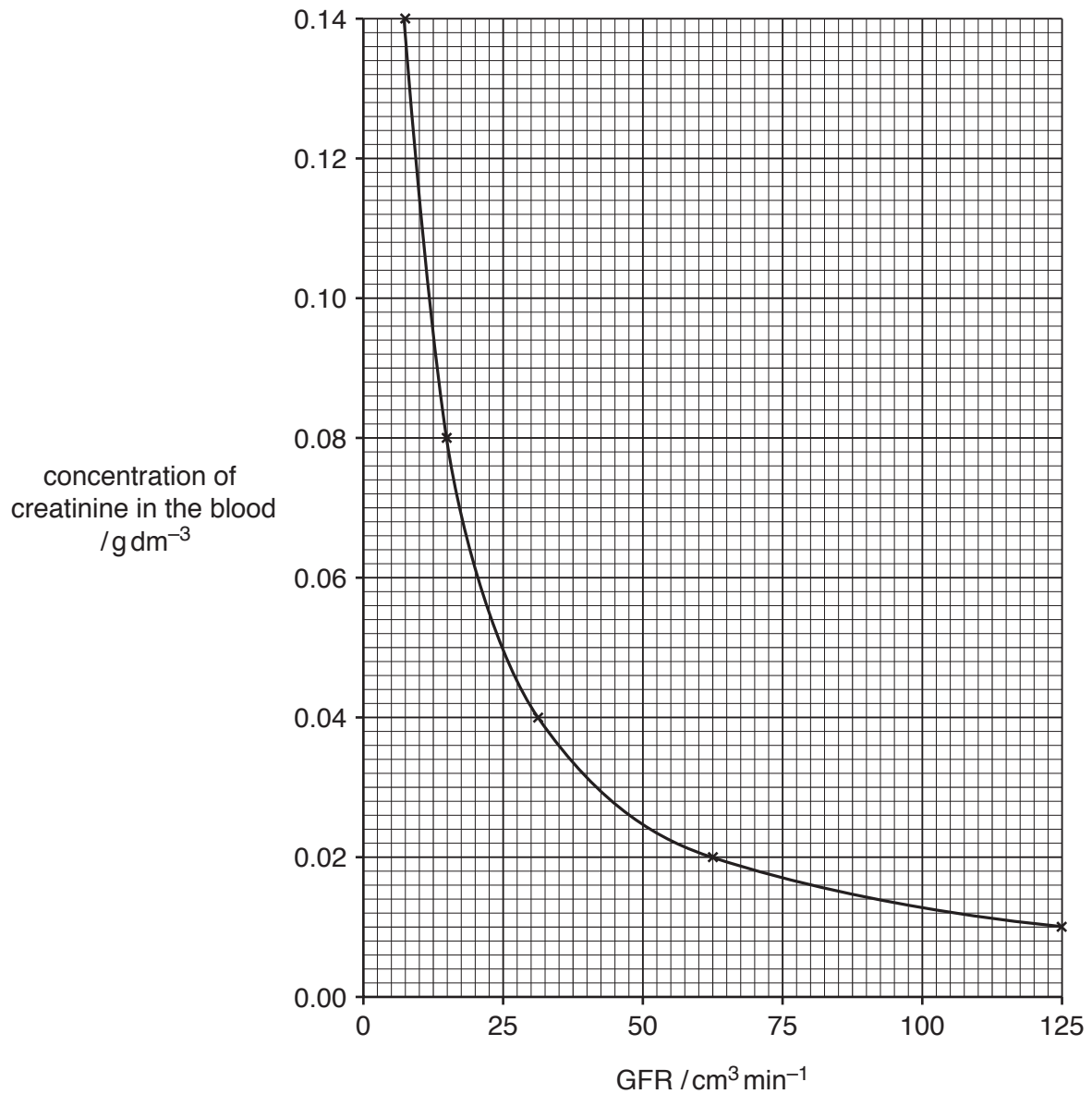


Fig. 5.2

(i) Describe the relationship shown in Fig. 5.2.

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(ii) Use Fig. 5.2 to estimate the concentration of creatinine in the blood that indicates a normal GFR.

answer[2]

(iii) Suggest **two** reasons why the GFR could decrease.

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[Total: 12]